

CoExplorer: Generative AI Powered 2D and 3D Adaptive Interfaces to Support Intentionality in Video Meetings

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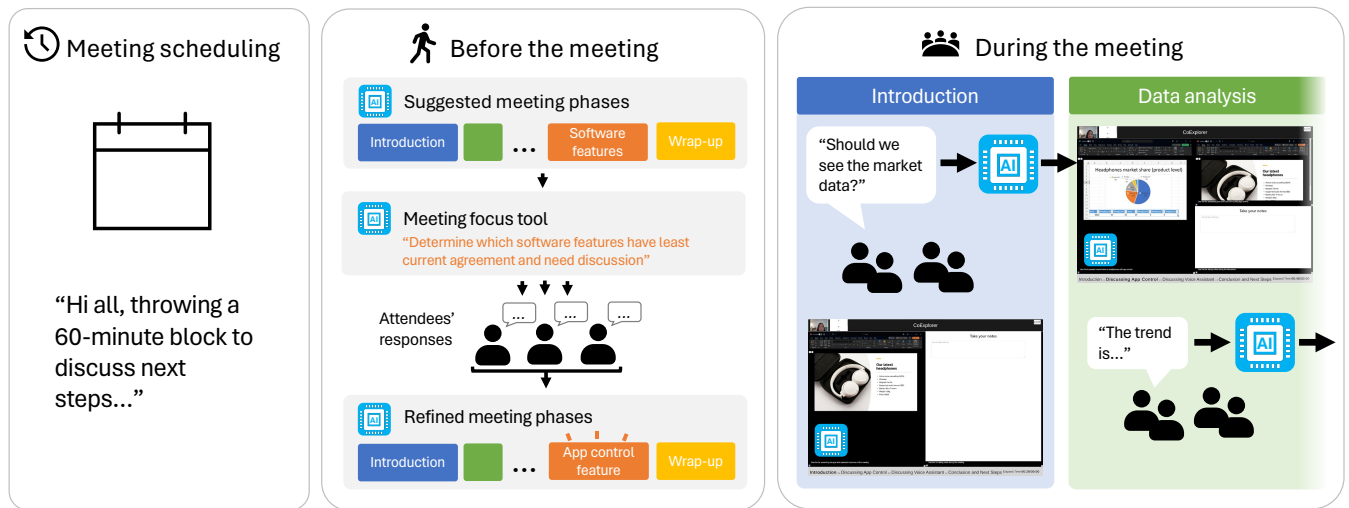


Figure 1: CoExplorer uses Generative AI (GenAI) to simplify the process of achieving meeting objectives. Before the meeting, a GenAI system analyzes an email meeting description and then generates proposed meeting phases as well as a tool for all attendees to identify discussion focus, from which the phases are further refined. In the meeting, the system monitors talk and facilitates transitions between phases, generating applications and optimizing the UI layout to suit each meeting phase.

ABSTRACT

Current online meeting technologies lack holistic support for reducing the effort of planning and running meetings. We present CoExplorer2D and CoExplorerVR, generative AI (GenAI)-driven technology probes for exploring the significant transformative potential of GenAI to augment these aspects of meetings. In each system, before the meeting, these systems generate tools that allow

synthesis and ranking of attendees' key issues for discussion, and likely phases that a meeting would require to cover these issues. During the meeting, these systems use speech recognition to generate 2D or VR window layouts with appropriate applications and files for each phase, and recognize the attendees' progress through the meeting's phases. We argue that these probes show the potential of GenAI to contribute to reducing the effort required for planning and running meetings, providing participants with a more engaging and effective meeting experiences.

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CCS CONCEPTS

• **Human-centered computing** → *Collaborative and social computing systems and tools.*

KEYWORDS

meetings, effectiveness, goals, planning, facilitation, videoconferencing, virtual reality, adaptive user interface, windowing, speech recognition, intent recognition, design, generative AI

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1 INTRODUCTION

Productive meetings require effort before, during, and afterwards [64, 73]. However, many organizations neglect formal training for meetings [3], and meeting systems themselves do not provide holistic support for reducing the effort of meeting planning and execution. Generative AI (GenAI) could be transformative in reducing the effort needed to conduct effective meetings. Existing commercial videoconferencing platforms such as Zoom Companion AI [45] and Microsoft Teams Meeting Copilot [55] are already incorporating GenAI capabilities to improve meetings. Nonetheless, we posit that the scope of GenAI's benefits to meeting effectiveness extends far beyond current implementations.

This paper introduces CoExplorer, a technology probe designed to foreground intentionality in distributed meeting systems and challenge classic conceptions of the meeting stage. A shared GenAI infrastructure underpins two interfaces: CoExplorer2D for traditional videoconferencing and CoExplorerVR for virtual reality meetings. CoExplorer uses natural language from a meeting invitation to (a) predict potential meeting objectives and structure, (b) craft a tool to maintain focus on the presumed meeting objectives, and (c) recommend a tailored set of spatially configured applications for distinct meeting phases. CoExplorer autonomously refines the phase outline (i.e., meeting structure) by assessing the context of ongoing tasks, modulating the interactive content. CoExplorer2D assumes a need to fit into traditional 2D meeting interfaces, but uses a dynamic windowing system to adapt content to meeting phases. CoExplorerVR looks forward to 3D virtual reality meetings, and probes new design prospects for the dynamic display of content incorporated into the spatial context. The contributions of this paper are as follows:

- We introduce CoExplorer2D and CoExplorerVR, provocative designs for GenAI-driven meeting systems that adapt the working space in line with meeting goals and contextual activities, and reflect on key opportunities and challenges of adaptive meeting interfaces that support intentionality.
- We provide example prompts that could facilitate design research on adaptive GenAI-driven meeting interfaces.

2 RELATED WORK

Effectiveness of Meetings: Effort is required when planning for effective meetings [17, 33, 58, 77]. Setting goals requires significant time and effort [8, 10, 31]. Lack of understanding of appropriate tasks for meetings is also common [9, 52, 54, 69], and time pressures often lead to inadequate meeting preparation [26, 32]. Running and

participating in effective meetings also require effort. Agendas are valuable in structuring meetings [10, 19]. Managing the agenda while allowing flexibility for new ideas and disagreements is important [42, 43, 73, 86]. Determining which topics should be discussed during the meeting versus asynchronously resolved beforehand can reduce agenda length and create space for productive discussion [30, 57, 69]. Additionally, managing a meeting involves shepherding its phases [2], which may align with or go beyond agenda items, and ensuring participants are focused on relevant materials at the appropriate time [23]. Post-meeting effort involves capturing action items or next steps [41, 59, 61], and then acting on them, but if a meeting lacks clear goals or is poorly conducted, post-meeting effort is increased [73]. The effort required for meetings applies to both in-person and online meetings. However, online meetings introduce additional effort.

2D and VR Meetings: The ubiquity of 2D videoconferencing is evidenced by platforms like Microsoft Teams and Zoom. The meeting stage of most commercial systems has been largely unchanged since the 1990s, generally consisting of a grid of people or strip of people adjacent to a single pane of content. The COVID-19 pandemic showed that constant exposure to this stage is not optimized for meeting effectiveness and is also fatiguing [7, 9, 47, 72]. 2D virtual worlds (e.g. Gather [1]) introduce persistent spatial environments that hold some promises for curbing fatigue by virtue of variety of moving into and out of meetings. However, once actually in a meeting, they revert to traditional A/V elements, they make no direct changes to planning or running the meeting, and little change to factors of fatigue [46, 50, 66]. The use of 3D Virtual Reality (VR) meetings is slowly increasing (e.g. in Meta Horizon Workrooms [53], Microsoft Mesh [56]). These VR meetings improve on the sense of person and spatial presence of traditional 2D stages, potentially enhancing overall conversational flow as well as fluid movement between the full meeting and smaller huddles [48, 60, 62, 79]. However, the extent to which VR can be designed to enhance meeting effectiveness is not yet well understood.

Adaptive User Interfaces: Adaptive UIs morph based on user requirements, either during initial setup or throughout their operation, streamlining user efficiency and reducing cognitive strain [4, 12, 27, 28, 51, 78, 85]. Solutions like adaptive windowing systems facilitate immediate access to necessary tools relevant to the task at hand [36, 67, 80, 83, 84]. However, within meetings' specific phases and task-related content, the adoption of adaptive strategies remains scarce. Genre analysis suggests that meeting types could inform tailored UI design, but issues of scale indicated a need for generative AI's capabilities [5, 16].

The spatial environment in 3D VR (and Augmented Reality/physical reality) contexts provides specific contextual cues based on the user's position. For example, standing next to a cooktop implies that this area is intended for interactions related to heating a pot or pan. Although there has been some limited exploration on using these contextual cues to inform the adaptation of placement of VR interactional and informational elements [15, 70], little research has examined how the system can automatically adapt these spatial placements as the meeting progresses.

textbfUsing AI to Prototype Interfaces for Improving Meeting Effectiveness: AI offers tools to streamline the various stages of meetings, including pre-meeting scheduling [17] and agenda item

voting [10, 29, 30], decision-making support during the meeting [22, 39], and post-meeting tasks such as summary generation and action item tracking [6, 41, 59, 61, 74] (although we do not focus on post-meeting issues in this paper). Prototyping with LLMs such as GPT-3 simplifies the development of decision-making algorithms, benefiting from their capabilities to process natural language with minimal data [11]. LLMs combined with proactive Voice User Interfaces (VUIs) can provide real-time support, aligning meeting dialogue with predefined objectives [21, 75, 87, 89]. Despite the non-deterministic nature of LLM outcomes, exemplified by ChatGPT's unpredictable predictions, their application in commercial systems suggests a promising avenue for AI enhancement in meeting contexts [14, 25, 35, 40, 49, 68, 88].

In this exploration, we focus on how to design system capabilities that *contribute* to meeting effectiveness. These include reducing struggles of meeting *practice* such as improving pre-meeting articulation of meeting goals and getting attendee buy-in to the focus for agenda items, as well reducing struggles with meeting *technology*, such as making the the right resources available at the right time and in the right place for all attendees for different meeting phases. More direct evaluation on the effect these capabilities have on meeting effectiveness remains for future work.

3 DESIGNING COEXPLORER

Through internal discussion based on our own work experience and prior research, from a large initial set of scenarios we chose one **meeting scenario** that (a) would be familiar for our knowledge worker participants (and, at a high level, knowledge workers more generally), and that (b) would benefit from GenAI-augmented help with planning and running a meeting. We landed on meetings of cross-functional product teams in the technology industry, which often face struggles of coordination [38], especially when planning a course of action about which they have different opinions and stakes, and for which the resources are scattered across different storage locations and apps.

We developed a fictional scenario centered on the need for effective decision-making. In this scenario, a team's current product, the "Strata Headphones 2", is lagging behind competitors in market share, and so the team needs to decide on the features necessary to increase market share of the forthcoming "Strata Headphones 3". The team has files such as the list of design, hardware, and software features, their cost, competitor product information, and the current specification sheet. We used this scenario to formulate three design strategies.

3.1 Design Approaches

Design Approach 1: Incorporate collective feedback to shape meeting objectives. An agenda is typically set by the person organizing the meeting, but should also reflect the priorities of other attendees [82], yet current videoconferencing systems do not easily allow for collaborative crafting of meeting objectives. Participants' contributions can be integrated through mechanisms like voting systems to create or fine-tune the agenda [30]. This design approach explores the potential of a GenAI system to produce a tool that can assimilate varying viewpoints, focusing the conversation on reconciling differences.

Design Approach 2: Clarify underlying needs and available resources. The clear definition of objectives, agendas, preparatory materials, and task-related resources is an essential aspect of efficient meeting preparation and facilitation, notably in virtual settings [18, 44]. Yet, due to time constraints, both meeting planners and participants may cut corners in laying out pre-meeting preparations [76]. During the meeting, while *explicit* items on the agenda direct the discourse, the meeting actually often transitions through *implicit* phases. These phases may align with agenda points but can also encompass several points or be subdivided into finer details. Moreover, essential files and applications are often linked to each action-oriented phase. The implicit nature of these phases makes accessing appropriate resources at the opportune moment burdensome. This design approach explores the challenges and potential benefits of GenAI in identifying implicit phases as they arise during the meeting, and using them to drive a specific arrangement of resources and the meeting's intended progression.

Design Approach 3: Manage the system through a Human-on-the-Loop (HOTL) method. HOTL characterizes human-machine interactions in which the automated system mainly allows humans to *abort* the machine's decisions [63]. This grants automated systems more independence and limits the number of prompts to users, which is compatible with a meeting context where participants often have lower capacity to process information outside the interpersonal interactions of meetings [65]. Introducing HOTL in meeting systems has been impeded by inadequate predictive performance in natural language models, but with the launch of GPT3.5, there has been noticeable improvement in accuracy across a spectrum of fields [14, 40]. Nonetheless, to foster and sustain the trust of users, HOTL systems must clear a higher threshold than the often-preferred human-*in-the-loop* methodology in Human-Computer Interaction [24, 68, 88] [63]. This design approach explores the difficulties and possibilities of building trust with users through HOTL, and how this might influence user perceptions of both efficiency and adaptability.

3.2 Overall implementation:

We implemented CoExplorer using Unity and GPT-3.5. For the real-time video communication (only available for CoExplorer2D), LiveKit was utilized. This choice was made to provide as much cross-compatibility as possible. CoExplorer2D runs on standard Windows computers, while CoExplorerVR runs on Meta Quest 2. See Appendix A for versions of example prompts used with GPT-3.5 to power CoExplorer. In the sections below we detail aspects of the CoExplorer2D and CoExplorer3D systems.

3.3 How CoExplorer2D Facilitates Meetings

3.3.1 Formulating the Initial Phases of the Meeting. The meeting organizer distributes a meeting invitation, and as the meeting time approaches, for each attendee CoExplorer2D outlines the meeting's aim (Figure 3 (A), top left) and describes the basis of that aim (Figure 3A), bottom left). It also suggests phases anticipated for the meeting (Figure 3A), left, shown as segmented bars). Details for each segment include the name of the phase, the expected duration, and the pertinent activities for that phase. Below we expand on how the system implements the design strategies above. The strategies

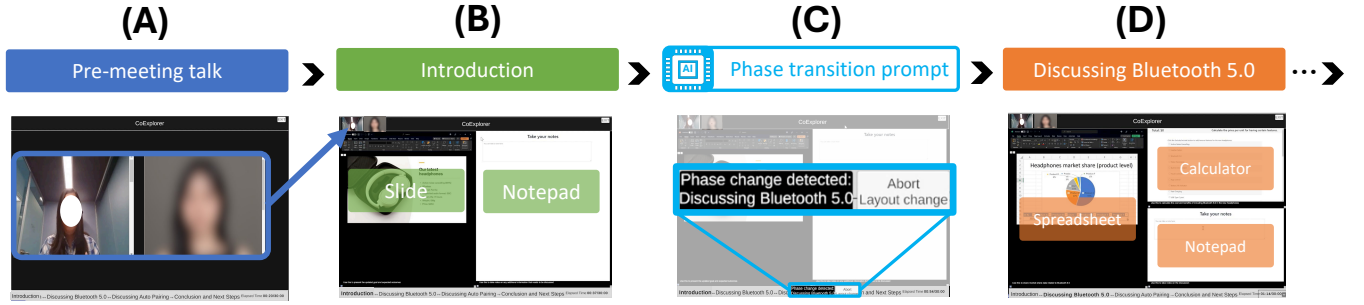


Figure 2: CoExplorer2D listens for meeting attendees transitioning between phases, then notifies the attendees that a change has been detected and modifies the layout to accommodate the new phase.

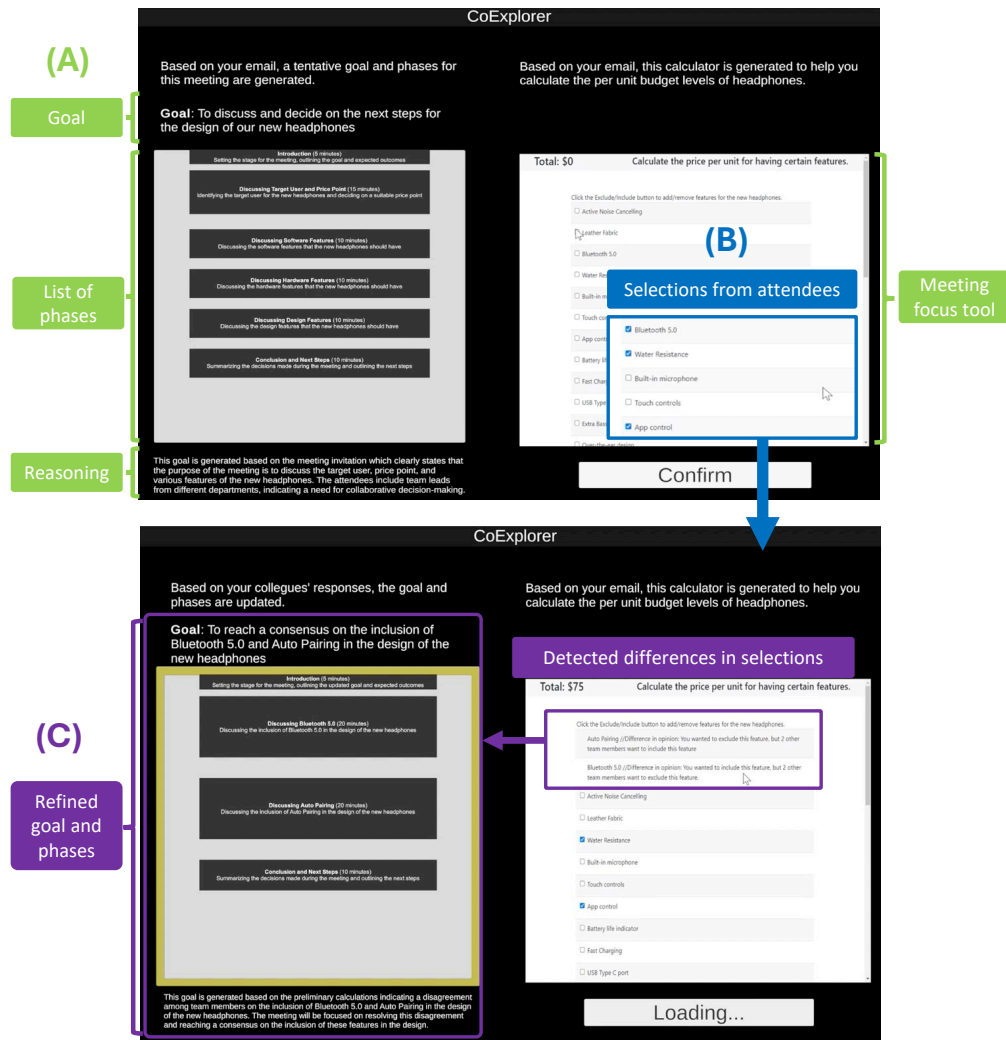


Figure 3: (A) Sequence of phases and tool for determining meeting focus. (B) Meeting attendees employ the focus tool to select preferred features. (C) CoExplorer uses the aggregated preferences to adjust the meeting's objective and phases.

are not implemented in the order presented, and one strategy might be implemented multiple times.

Implementing Design Approach 2: Inspired by the successful use of chaining in Large Language Models, we posited that, given a brief meeting invitation text, GenAI could identify both the purpose of the meeting, and a tailored list of applications for each phase. We emphasize our focus is on exploring the potential applications enabled by GenAI’s capabilities, and thus formal evaluation of this capability is out of scope for this paper. GenAI is tasked with elaborating on the provided invitation text and generating a list of phases of the meeting (see Appendix A), complete with titles and descriptions. We then guide GenAI to structure these details into a coherent list that includes explanations for the choices it has made, crafting a narrative that can both inform the CoExplorer2D system and persuade users of the validity of the decisions. To achieve this, we initialise GenAI requests with system prompts, bolstered by examples, to ensure the output is clear and actionable. This preliminary information generated by GenAI is fixed at the beginning and does not undergo refinement during the meeting, intentionally designed to preclude any confusion that might arise from changes during the meeting.

Implementing Design Approach 1: CoExplorer2D also primes participants to consider their own requirements and those of the team by generating a discussion initiator. In this scenario, the discussion initiator takes the form of a Meeting Focus Tool that lets meeting participants assess the implications of including specific features in a product from their role perspective and voice their preferences on key features. This tool is displayed on the right-hand side of Figure 3B as CoExplorer2D creates it. Once all preferences are communicated, as seen in Figure 3B, CoExplorer2D synthesizes these varied responses to refine both the meeting’s objectives and phases. CoExplorer2D presents the revised information as exhibited in Figure 3C, which narrows the team’s focal points for discussion to areas with the most divergent views. Areas of divergence are chosen for discussion because live meetings are most suited to dynamic discussions, including productive conflict, while areas of agreement can be handled asynchronously or set aside for later [81]. In this scenario, the GenAI system proposed that the tool take the form of a feature ranking aid (Figure 5). GenAI’s versatility would also enable a meeting focus tool to manifest in a range of alternative ways, such as a chatbot or a visualization.

During our experiments, we observed that GenAI is indeed proficient at devising a tool for this need. To enable a later evaluation, we opted to employ a pre-generated version of the tool rather than generating it on-the-fly. This decision was made with the intention of ensuring a consistent experience for all users. Upon utilizing the pre-generated tool, we noted certain aspects of the user interface, such as the color scheme used for highlighting selected buttons, could be misinterpreted. Consequently, we implemented manual adjustments to the pre-generated version to refine the prototype for better clarity and functionality.

3.3.2 Dynamic Window Management. Upon examination of the updated phases, the team commences the meeting. CoExplorer2D curates the necessary documents and applications for each phase, generating an ideal layout for the display. Initially, participants engage in a social introduction phase, where their video feeds are

maximized (Figure 2A). As the meeting progresses from casual conversation to its formal agenda, CoExplorer2D detects the shift to a new phase, specifically the project introduction. The project manager outlines the problem being addressed.

Implementing Design Approach 2: To process spoken dialogue, we segmented speech into discrete utterances based on pauses. For each utterance, a transcript was produced using the Microsoft Azure Speech API. These transcripts were then provided to GenAI, which determined the pre-identified phase of the meeting the utterance pertained to. If GenAI’s prediction indicated a new phase that differed from CoExplorer2D’s current phase, the user interface was updated to reflect the new phase, shifting to the window layout associated with it. This layout switch was based on a pregenerated list of window configurations established at the start of the meeting. A small latency was introduced during this process, but since CoExplorer strives to proactively take action, rather than react to user prompts, this latency was not especially noticeable and not directly relevant to the tasks. We expect the latency to decrease as LLMs improve (e.g. see Mixtral 8x7B Instruct¹).

In organizing the layout of multiple windows, we employed a tiling approach. This decision was influenced by our experimentation with GenAI’s ability to generate window sizes and positions. While GPT-4 was adept at creating well-fitted window layouts, GPT-3.5 — the version available to us during prototyping — fell short in this capacity. Consequently, a freeform window layout was deemed unsuitable. We established that a tiling window layout optimally utilized the available screen space within CoExplorer. Tiling thus became the chosen method, ensuring a high degree of screen real-estate utilization across the interface in our prototype.

Implementing Design Approach 3: At this juncture, CoExplorer2D asks all participants on whether they want to halt the transition (Figure 2C). If no objections occur, CoExplorer2D adapts the screen layout to fit this more work-focused phase: video sizes are reduced, with a PowerPoint presentation occupying the left side of the display, and a collaborative notepad on the right (Figure 2B).

In the meeting storyline, the hardware engineer begins discussing the importance of Bluetooth 5.0 for the headphones, picked up on by a software engineer who points out the challenges in supporting this feature through software. Once more, CoExplorer2D senses a phase shift (requesting a confirmation on whether to proceed with the change (Figure 2C). With no opposition, CoExplorer2D transitions to a “Discussing Bluetooth 5.0” phase (Figure 2D). The PowerPoint is replaced with an Excel sheet on the left side, a calculator for the Meeting Focus Tool atop the right side, and the notepad downsized to the bottom right, guiding participants back to the highlighted contentious topic.

A full meeting would proceed this way to a decision and discussion of next steps. Due to time limitations, we did not complete the full meeting scenario in CoExplorer2D. Current Generative AI systems such as Microsoft Copilot are able to detect and outline action items from a transcript, and thus a future CoExplorer-like system could use this in conjunction with adaptive windowing to place action items under the video of the relevant person.

¹<https://artificialanalysis.ai/models/mixtral-8x7b-instruct>

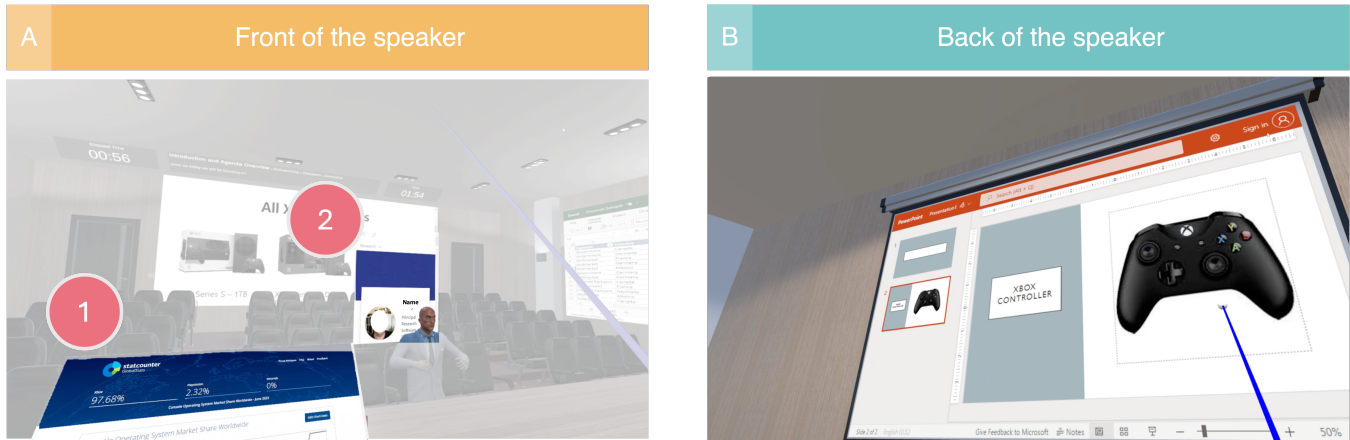


Figure 4: Views using CoExplorerVR. (A) Presenter's view of the audience; (B) Audience's view of the presenter's content.

3.4 Enhanced Spatial Interaction in CoExplorerVR

CoExplorerVR integrates spatial context into its support for seamless meeting facilitation, enhancing the interactions in a 3D virtual environment as depicted in the meeting life cycle (Figure 2). Spatial context refers to the significance attributed to specific locations within the 3D scene [15, 70]. The richer context means that window placement in CoExplorerVR needs to integrate with semantic relevance to the virtual environment more than window positions do in CoExplorer2D. In contrast to the CoExplorer2D, where GenAI optimized window positioning and sizing on a single flat plane, CoExplorerVR expands this functionality into the third dimension, allocating virtual windows to appear around a room in meaningful spatial arrangements.

Visibility: Visibility in a 3D space is a nuanced aspect, as certain objects or windows may only be visible from specific perspectives. A window placed on the main speaker's table (Figure 4A-1) is primarily intended for personal viewing, given the constraints of perspective and field of view. Conversely, a large display positioned behind the speaker (Figure 4B), is intended to be easily visible to the audience.

Object-associated context: In a 3D setting, objects can carry inherent meanings based on their function or placement. CoExplorerVR capitalizes on this by associating contextual information with objects, such as situating a digital contact card at the site of an audience member's chair (Figure 4A-2).

Controllability: The interactivity of objects and information panels in VR is critical. The location of the contact card allows for easy access, yet it also occludes the panel it overlaps, posing a challenge for the ray-based pointer typically used in VR for selection and control (Figure 4A-2). This aspect of spatial controllability must be carefully managed to ensure user comfort and efficiency within the virtual environment.

By incorporating these contextual parameters, the GenAI receives a comprehensive system prompt enabling intelligent window placement that respects the spatial context (given as set of tiles with associated description of the context) and aligns with meeting progression, in parallel to CoExplorer2D's approach.

4 FUTURE WORK

Future Development: GenAI drastically cuts the time and data requirements for developing systems like CoExplorer by eliminating the usual steps of data collection, model training, and testing—instead needing only a few examples. However, validating the outputs from GenAI remains critical for future development. Key challenges include the following:

- **False Positives:** GenAI tends to generate non-null, extensive-change responses due to its sensitivity, leading to inappropriate output. This is exacerbated by training on human-evaluated datasets with inherent errors [13].
- **Processing Time:** Real-time utterance processing is impeded by the slower speeds of Large Language Models (LLMs) compared to current speech-to-text systems.
- **2D/3D Windowing:** Both 2D and 3D display systems offer unique potential for GenAI to enhance user experiences. 2D interfaces benefit from the flexibility in window placement and management, albeit presenting the challenge of handling overlaps and layout dynamics. Our tests found GPT-3.5 struggling with screen space optimization, implying a need for more sophisticated solutions. For 3D displays, adapting to physical contexts for window allocation is critical, requiring further progress in multimodal models.
- **SDK Transition Hurdles:** LiveKit enables real-time video call implementation, but its lack of support for Unity clients led to workarounds. This is particularly problematic for real-time communication due to the critical need for low-latency performance. A more flexible SDK could mitigate these compatibility issues.

Future Evaluation: Our plan is to evaluate CoExplorer2D using a technology probe-based approach [37, 71], which will provide essential insights into the unique opportunities and challenges presented by GenAI-supported meeting systems. Further evaluation of how the ideas in these technology probes might directly impact meeting effectiveness should be conducted using methods similar to that in past work, which includes surveys and telemetry [20, 34].

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A APPENDIX

We attach versions of the example prompts for GPT 3.5-Turbo that we used for implementing CoExplorer. CoExplorer generated prompts automatically based on the templates. During this, CoExplorer optimized the prompts as per the user inputs, and prompted the language model multiple times to obtain the results in the correct format. The prompts given is one instantiation of output prompts that the CoExplorer was producing.

A.1 Phase Generation

A.1.1 Example system prompt.

[No prose]

[Output only JSON]

Do not write normal text.

You are a JSON generator which converts meeting agenda text into a more descriptive agenda description.

You always need to have an introduction phase at the beginning.

A.1.2 Example user prompt.

Please break down the following meeting agenda that someone has sent in email into meeting phases that we would need to go through.

We have a 60 minute meeting scheduled.

Based on the given information, give a goal of the meeting (goal), as well as the explanation on why you chose the goal (exp).

And also give the phase definition in a list (pi).

Each phase definition should include: (1) Phase title (pt), (2) Phase description (pd) which should include a sub goal of a phase, (3) Behaviors to be encouraged (be), (4) Behaviors to be discouraged (bd), (5) priority (p), (6) amount of time allocation (t), (7) direction (d) (i.e., is it an iterative phase or directional phase).

Please only respond in JSON with each element needed as a key within a phase,

i.e., if we have two phases,

```
[{"pt": "xxx", "pd": "xxx",
  "be": ["xxx", "yyy"], "bd": ["xxx", "yyy"],
  "p": "high", "t": 2, "d": "iterative"},
 {"pt": "yyy", "pd": "zzz", "be": ["ttt", "kk"],
  "bd": ["lll", "mmm"],
  "p": "low", "t": 8, "d": "directional"}].
```

So the overall JSON to export is {"goal": "xxx", "pi": [
<phase definitions>], "exp": "xxx"}.

Explanation should start by saying "this goal is generated..." or similar.

Please use the full 60 minutes.

Here is the meeting invitation:

(The email invitation would be attached.)

Refinement could be done using the same script, attaching the refinement scenario. This script can be generated through code, and does not have to be detailed.

A.2 Layout Generation

A.2.1 Example system prompt.

[No prose]

[Output only JSON]

Do not write normal text.

You are a helpful assistant who creates screen layout that has appropriate apps that are most helpful for users to complete the task successfully.

Respond only in JSON following the format.

Example format:

```
[{"PhaseTitle": "xxx", "timer": n, "programList":
 [{"name": "yyy", "description": "zzz"},
 {"name": "kk", "description": "lll"}]},
 {"PhaseTitle": "xxx", "timer": n, "programList":
 [{"name": "yyy", "description": "zzz"},
 {"name": "kk", "description": "lll"}]}].
```

Strictly follow this format.

n is integer, and programList.name should either be a name of a program in a program list given or a URL.

programList.description is where you should put

an extremely concise reason why you chose that program.

e.g., Use this for presenting agenda;

Use this for viewing relevant budget data

A.2.2 Example user prompt.

I will give you the list of phases in a meeting in JSON format.

Each phase in JSON is defined with several keywords.

"pt" represents phase title, "pd" represents phase description, "be" represents behaviours to be encouraged, "bd" represents the behaviours to be discouraged, "p" represents priority (high, medium, low), "t" represents recommended duration of time for the phase, and "d" represents directionality (directional i.e., cannot be returned, and should be preceded by a certain phase or iterative i.e., can be transitioned to this whenever). You need to generate what kind of programs are needed for helping goals of each phase (defined by the description) to be met the most efficiently.

You can generate a list of 1-5 program name/URL, and the sequence of generation will affect where they are being placed, and size.

Therefore, you need to be sensible about ordering so that important programs can be shown with the bigger sizes.

The rule is as follows:

If you have one program on the list, that would be full screen. If you have two programs, it would be one on the left half (first program on the list), and one on the right half.

If you have three programs, the right-hand-side panel will split in half, creating two small panels at the top and the bottom.

Four programs mean the left-hand-side panel will also be split. Five programs mean two equally sized panel at the top, and three equally sized panel at the bottom.

The ordering in the list will be used to place program to panels in a clockwise ordering (top left panel is the first panel).

Here is the list of programs available, and if the program that you want is not listed, please generate a URL for the program that you need instead of the program name.

Please feel free to give Bing Search URL with the search term filled, and generate at least one URL:

(List of programs could take arbitrary formats.)

A.3 Meeting Focus Tool

A.3.1 Example system prompt for a specialized calculator generation.

You need to generate a HTML page, and only HTML+CSS+JavaScript based page as a response which allows me to calculate the total value to have the a list of features for a given scenario.

Each feature needs to have an "include" button (green tick) or "exclude" button (red cross), and you need to calculate the total at the end, when the submit is clicked.

You should not show any prices including itemised ones before this.

You need to generate at least 30 features (features should be unique and descriptive. No "Feature 26" or something like that), and incorporate that into the page, those are relevant for the scenario.

Assign random prices for each feature.

The HTML page needs to incorporate all the features that you generated embedded (i.e., no "..." or "many more features here" etc.).

No prose.

No add more features here etc.

You need to list all the features on the HTML.

A.3.2 Example user prompt.

A designer, software engineer, hardware engineer, PM, marketing expert, and a researcher are gathering to think about what features that a new headphone product that they release might have.

It could be electronics feature such as active noise cancelling (95\%) or it could be about materials (leather etc.).

Rather than listing explicit features, this user prompt may occasionally result in "*<!--Here will be inserted many more features-->*". Such cases can be detected, and additional user prompt could be given to force the system to be explicit by pointing out that it has not followed instruction. The prompt could take a form as follows:

It does not show 30+ features

The system would then generate a web page as in Figure 5:

Calculate

Total: \$955

- Active Noise Cancelling ☐ Exclude
- Leather Fabric ☒ Include
- Bluetooth 5.0 ☒ Include
- Water Resistance ☒ Include
- Built-in microphone ☒ Include
- Touch controls ☒ Include
- App control ☒ Include

Figure 5: Example raw web app generated by GPT 3.5